

Computer Vision - 2026

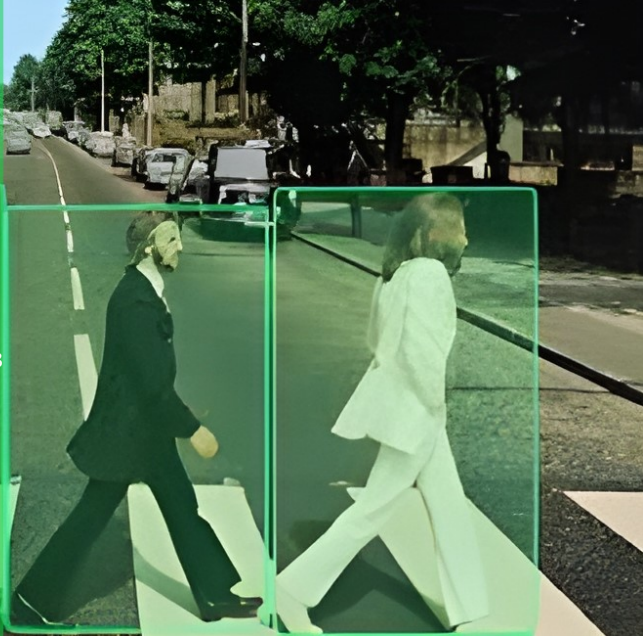
Week #12. Vision - Language - Action Models

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Agenda

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Modern VLA
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Training
Modern VLA
Systems

Modern VLA
Post-Training
with
Reinforcement
Learning

① Modern VLA Architecture

② Training Modern VLA Systems

③ Modern VLA Post-Training with Reinforcement Learning

From Perception to Action

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Computer vision traditionally focused on prediction:

Image $\rightarrow$ Label

Modern embodied systems require control:

Observation $\rightarrow$ Decision $\rightarrow$ Action $\rightarrow$ Feedback

The system must continuously interact with the environment.

- perception estimates the current state
- action modifies the environment
- feedback determines the next decision

This interaction loop defines intelligent behavior in physical systems.

Recognition Is Not Control

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Recognizing an object does not imply the ability to manipulate it.

Example:

- a model detects a cup
- the robot must decide:
 - where to grasp
 - how much force to apply
 - how to move safely

The central challenge becomes:

What action should be executed next?

This shift transforms vision systems into control systems.

What Is a Vision-Language-Action Model

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A Vision-Language-Action (VLA) model extends perception with decision-making.

Input:

$$s_t = (\text{visual observation, robot state, task instruction})$$

Output:

$$a_t = \text{control command}$$

The model predicts an action that changes the physical world.

- vision provides information
- language defines the goal
- action executes the plan

Closed-Loop Control

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Unlike image classification, robotic systems operate in a loop.

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$$s_t \rightarrow a_t \rightarrow s_{t+1} \rightarrow a_{t+1} \rightarrow \dots$$

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Each action changes the environment and produces new observations.

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- the system observes
- the system acts
- the environment responds
- the cycle repeats

Reliable behavior therefore requires continuous adaptation.

The Core Question of Embodied Intelligence

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Modern AI systems must operate in uncertain and dynamic environments.

The fundamental decision problem is:

$$a^* = \arg \max_a P(\text{success} \mid s, a)$$

The system selects the action most likely to achieve the goal.

- perception estimates the situation
- planning evaluates alternatives
- control executes the selected action
- feedback validates the result

Why Vision-Language-Action Models Matter

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VLA systems represent the next stage of AI development.

They enable machines to:

Capabilities

- manipulate objects
- follow natural language instructions
- adapt to changing environments
- operate autonomously

Applications

- household robots
- warehouse automation
- medical assistance systems
- industrial inspection

The focus shifts from prediction to reliable behavior.

Learning Outcomes

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After completing this lecture, students should be able to:

- **Explain** what a **Vision-Language-Action (VLA)** model is and how it differs from traditional perception systems
- **Describe** the modern VLA architecture pipeline:

observation → representation → policy → action

- **Identify** the main stages of training a modern robot policy:
 - foundation model pretraining
 - supervised imitation learning
 - reinforcement learning post-training
- **Interpret** the role of large-scale robot datasets (e.g., LIBERO, RoboTwin) in improving generalization
- **Analyze** how reinforcement learning can improve performance after supervised training
- **Evaluate** the difference between imitation and exploration in robot learning
- **Recognize** the emerging pattern of capability growth through interaction rather than manual supervision

Operational skill

Students should be able to read a modern VLA paper and reconstruct its training pipeline and evaluation logic.

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Section 1. Modern VLA Architecture

Modern VLA Architecture: The Main Question

A VLA maps multimodal context to executable control:

(image, robot state, instruction) $\rightarrow$ action

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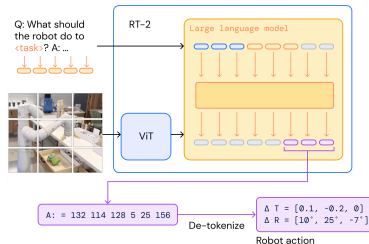
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Why this is harder than perception-only models

- integrates visual scene understanding with language goals
- estimates physical state and predicts robot-safe commands
- must adapt online from feedback after each action step

Core architecture choices

- **Encoder:** CNN / ViT / multimodal perception backbone
- **Fusion:** concatenation, cross-attention, autoregressive mixing
- **Action head:** continuous control, tokenized actions, or diffusion chunks
- **Rollout:** open-loop chunking vs. closed-loop replanning



Canonical VLA Pipeline

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Operational view

A modern VLA follows a four-stage mapping:

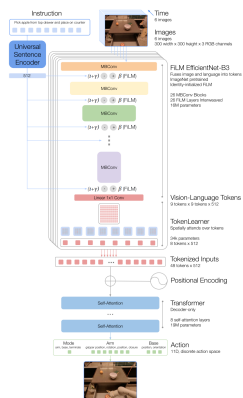
Perception Encoder $\rightarrow$ Multimodal Fusion $\rightarrow$ Policy Backbone

Policy Backbone $\rightarrow$ Action Decoder

Inputs and outputs

- **Inputs:** RGB image(s), optional depth/point cloud, proprioception, language goal, history
- **Outputs:** end-effector pose delta, joint action vector, gripper command, action chunk
- Action format may be continuous, tokenized, or denoised (diffusion-style)

This abstraction covers RT-1, RT-2, Octo, and OpenVLA [Brohan et al. \[2022, 2023\]](#); [Collaboration et al. \[2024\]](#); [Kim et al. \[2024\]](#).



Architecture Taxonomy: Three Main Families

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Family	Main Idea	Strength	Representative Models
Transformer BC policy	Autoregressive or sequence policy maps observations to actions	Simple scaling, unified token pipeline	RT-1 Brohan et al. [2022] , RT-2 Brohan et al. [2023] , OpenVLA Kim et al. [2024]
Diffusion action policy	Predict action chunk by iterative denoising	Smooth multimodal trajectories, robust low-level control	Diffusion Policy Chi et al. [2023] , RAD / RDT-style descendants
Generalist multi-embodiment policy	Train one policy on many robots, tasks, embodiments	Transfer, broader coverage, better data reuse	Octo Collaboration et al. [2024] , Open X-Embodiment Collaboration [2023] , π_0 Black et al. [2024a]

Practical interpretation

The field is converging to a common recipe:

foundation-style perception + language conditioning + robot policy head

while differing mainly in **how actions are represented and decoded**.

What Does the Model Actually Predict?

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Representation	Model output	Strengths	Trade-offs
Continuous control	$a_t \in \mathbb{R}^d$ (pose/joint/gripper)	direct execution; simple head	harder multimodal prediction; less stable for diverse trajectories
Action tokens	$a_t \rightarrow z_t \in \mathcal{V}_{\text{act}}$	LLM-compatible training; autoregressive decoding	quantization error; tokenizer design matters
Diffusion chunks	$(a_t, \dots, a_{t+H-1}) \sim p_\theta(\cdot s_t)$	smooth trajectories; strong low-level control; multimodality	slower decoding; more complex inference

Rule of thumb

Token policies are best for **foundation-model scaling**; diffusion policies for **fine motor control**; continuous heads for **small, efficient systems**.

This taxonomy appears repeatedly across recent VLA systems [Brohan et al. \[2022, 2023\]](#); [Chi et al. \[2023\]](#); [Kim et al. \[2024\]](#); [Black et al. \[2024a\]](#).

Two Practical Blueprints

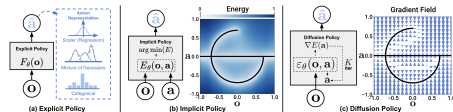
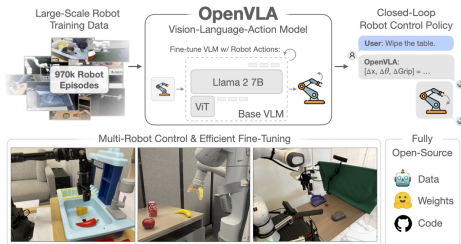
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B. Diffusion-based VLA

$(I_t, q_t, I) \rightarrow$ conditioned denoiser $\rightarrow$ action chunk

Typical for Diffusion Policy and related control models.

A. Token-based VLA

$(I_t, q_t, I) \rightarrow$ Transformer $\rightarrow$ action tokens $\rightarrow$ dequantize

Typical for RT-style and OpenVLA-style systems.

Architecture: Putting the Pieces Together

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Perception side

- RGB image encoder: CNN or ViT
- optional depth / point cloud encoder
- optional object-centric features
- temporal stacking or history tokens

Language side

- instruction tokens
- goal description
- sometimes dialogue or plan tokens
- may reuse VLM / LLM tokenizer and backbone

Fusion

- concatenate image + text + proprioception tokens
- cross-attention between modalities



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Visual decoding over mixed token stream

Policy side

- transformer backbone
- recurrent memory or history window
- predicts next action or action chunk

Execution side

- action decoder
- safety or clipping layer
- simulator / robot controller
- next state returned to the model

Design tension

A strong VLA must balance:

- general semantic knowledge
- spatial precision
- low-latency control
- robustness under distribution shift

Inference Algorithm for a Closed-Loop VLA

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Algorithm 1: Generic inference loop for a modern VLA policy

Input: Instruction I , initial observation o_0 , robot state q_0 , policy π_θ

Output: Executed trajectory τ

```
1 Initialize environment and set task goal
2  $\tau \leftarrow \emptyset$ 
3 for  $t = 0, 1, \dots, T - 1$  do
4     Encode visual observation  $f_t^{vis} \leftarrow E_{vis}(o_t)$ 
5     Encode robot state  $f_t^{prop} \leftarrow E_{prop}(q_t)$ 
6     Encode language instruction  $f^{lang} \leftarrow E_{lang}(I)$ 
7     Fuse multimodal context:
        
$$h_t \leftarrow \text{Fuse}(f_t^{vis}, f_t^{prop}, f^{lang}, \tau_{<t})$$

        Predict action representation:
        
$$z_t \leftarrow \pi_\theta(h_t)$$

        Decode executable action:
        
$$a_t \leftarrow \text{Decode}(z_t)$$

        Execute  $a_t$  in simulator / robot controller
8     Observe next state  $(o_{t+1}, q_{t+1})$ 
9     Append  $(o_t, q_t, a_t)$  to  $\tau$ 
10    if task success or termination then
11        | break
12    end
```

13 **end**

Representative Architectures

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Model	Backbone	Action Head	Main Message
RT-1 Brohan et al. [2022]	Token-based robot transformer	discretized action tokens	Scaling supervised robot policies works
RT-2 Brohan et al. [2023]	VLM adapted to robotic control	text/action token style decoding	Web knowledge can help action generation
Octo Collaboration et al. [2024]	Generalist transformer policy	action chunk prediction	Multi-embodiment data improves transfer
OpenVLA Kim et al. [2024]	Open vision-language-action backbone	7-DoF action prediction	Open-source foundation policy for manipulation
Diffusion Policy Chi et al. [2023]	Conditioned diffusion model	denoised action trajectory	Generative control is strong for manipulation
π_0 Black et al. [2024a]	Large-scale generalist robot policy	high-capacity action model	Scale and breadth are becoming central

Teaching note

Students do not need every detail here. They should learn to identify: **backbone, fusion strategy, action representation, rollout loop.**

Paper Reading 1: RT-1 and RT-2

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RT-1 and **RT-2** established the transformer line of VLA systems.

What to notice

- RT-1 treats robot control as sequence modeling
- RT-2 pushes further: a vision-language model is adapted to output robot actions
- the key conceptual shift is that semantic knowledge from large pretraining may help embodied control

Reading focus

While reading, identify:

- how actions are tokenized
- what enters the transformer context
- whether the model predicts one action or a chunk
- what kind of scaling claim the paper makes



Suggested reading: [RT-1: Robotics Transformer for Real-World Control at Scale](#) and [RT-2: Vision-Language-Action Models](#)
[Web Knowledge to Robotic Control.](#)

Paper Reading 2: OpenVLA and Octo

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OpenVLA and **Octo** represent the open, generalist direction of the field.

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OpenVLA

What to notice:

- open implementation matters
- VLA becomes easier to study in labs
- action generation is integrated with vision-language backbone design

Octo

What to notice:

- cross-embodiment training
- generalist policy viewpoint
- data standardization is part of the architecture story

Main lesson

Architecture is no longer only about layers. It also includes the **training interface**, **data format**, and **robot embodiment abstraction**.

Suggested reading: [OpenVLA: An Open Vision-Language-Action Model](#) and [Octo: An Open-Source Generalist Robot Policy](#).

Paper Reading 3: Diffusion Policy

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Diffusion Policy is the cleanest example of a generative control architecture.

Why it matters

It shows that robot action prediction is not only a classification or regression problem. It can also be treated as **trajectory generation**.

What to notice while reading

- actions are predicted as a short future sequence
- denoising models naturally represent multimodal solutions
- the architecture is often easier to connect to manipulation than large VLM-style token pipelines

Course connection

This is the direct bridge from your earlier lecture on diffusion / generative modeling to embodied control.

Suggested reading: [Diffusion Policy: Visuomotor Policy Learning via Action Diffusion](#).

Main Takeaways of the Architecture Section

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Three messages

- 1 A VLA architecture is a **closed-loop control system**, not only a multimodal predictor.
- 2 The main architectural choice is usually the **action representation**: continuous, tokenized, or diffusion-based.
- 3 Recent models differ less in high-level pipeline structure than in **scaling strategy, data regime, and action decoding**.

Compact summary

Perception + Language + Robot State → Policy Backbone → Executable Action then repeat in a loop

Once the architecture is defined, the next question is:

How do we train such systems well?

This leads naturally to imitation learning, action chunking, scaling data, and RL refinement.

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Section 2. Training Modern VLA Systems

From Models to Policies

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Modern VLA systems are not trained once.
They are trained in stages.



Modern training stack

- ① foundation model pretraining
- ② multi-robot data mixture training
- ③ task-specific fine-tuning

The Modern VLA Training Objective

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A VLA model learns a policy from demonstrations.

$$\pi(a \mid s, g)$$

where:

- s — observation
- g — instruction
- a — action trajectory

Training data therefore consists of:

(image, language, trajectory)

Large-scale robot datasets pair observations and instructions with trajectories collected from teleoperation or simulation.

Modern VLA Datasets

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Core datasets

- Open X-Embodiment
- BridgeData V2
- LIBERO simulation benchmark
- LeRobot community dataset

OpenVLA Training Pipeline

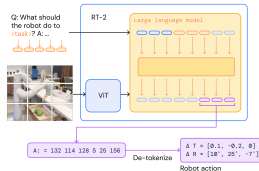
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Data scale

- 970k robot demonstrations
- multiple robot embodiments
- multi-task manipulation

OpenVLA uses a pretrained language model and visual encoder and is fine-tuned on diverse robot demonstrations.



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The $\pi_{0.5}$ Training Strategy

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Core idea

co-training on heterogeneous data

- robot demonstrations
- language data
- perception tasks
- semantic prediction

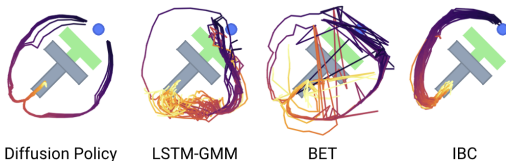


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intelligence [2025]

combines multiple data sources to improve generalization to new environments.

Flow-Based Policy Training

Modern VLA policies generate trajectories rather than discrete actions.



Modern control models

- diffusion policies
- flow matching policies
- continuous trajectory models

Continuous-output policies improve dexterity and control frequency in robotic manipulation.

Training Algorithm: Modern VLA

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Algorithm 2: Modern VLA training loop

Input: dataset of demonstrations

Output: trained VLA policy

- 1 Load pretrained model
 - 2 Initialize policy parameters
 - 3 **for each training step do**
 - 4 **end**
 - 5 Sample trajectory batch
 - 6 Predict action sequence
 - 7 Compute loss
 - 8 Update parameters
 - 9 Evaluate policy in simulation
 - 10 Deploy to robot
-

Paper Reading: Modern VLA Training

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Suggested reading

- OpenVLA (2024)
- $\pi_{0.5}$ (2025)
- PixelVLA (2025)

What to notice

- data scale
- training stages
- generalization strategy
- dataset mixture

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Section 3. Modern VLA Post-Training with Reinforcement Learning

Why Post-Training for VLA?

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Modern VLAs already have strong perception and policy backbones, but scaling pure supervised fine-tuning (SFT) is difficult.

Two main bottlenecks

- **Data scarcity:** large numbers of human teleoperation trajectories are expensive
- **Poor generalization:** SFT policies degrade under unseen tasks, scenes, and objects

Modern idea

Do not redesign the whole architecture. Take a strong pretrained VLA and **improve it by online RL** in simulation.

figures/simplevla_rl_overview.png

From SFT to RL Post-Training

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figures/simplevla_rl_pipeline.png

RL Formulation for VLA

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State

$$s_t = (o_t^{vis}, o_t^{prop}, l_{task})$$

- visual observation
- proprioception
- language instruction

Action

$$a_t \in \mathbb{R}^d$$

or tokenized / chunked action output.

- end-effector delta
- joint target
- gripper signal
- action chunk

Environment

The robot acts in a simulator:

$$s_{t+1} = \text{Env}(s_t, a_t)$$

Each action changes the observation.

Rollout

The policy repeatedly:

$$s_t \rightarrow a_t \rightarrow s_{t+1}$$

This is closed-loop control, not one-shot prediction.



The paper formalizes RL for VLAs exactly in this multimodal form, with iterative trajectory rollout in the environment ?.

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SimpleVLA-RL: Core Mechanism

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figures/simplevla_rl_core.png

Algorithm: Online RL Post-Training for a VLA

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Algorithm 3: Lecture reconstruction of the SimpleVLA-RL post-training loop

Input: Pretrained VLA policy π_θ , task batch $\mathcal{D}$, simulator Env

Output: Post-trained policy

```
1 for each task input in  $\mathcal{D}$  do
2   sample  $G$  trajectories by stochastic rollout from  $\pi_\theta$ 
3   execute trajectories in simulation
4   compute binary outcome reward for each trajectory
5   keep groups with mixed success / failure outcomes
6   estimate group-relative advantages
7   update policy with GRPO-style clipped objective
8 end
```

Interpretation

This algorithm is simple by design: no dense reward shaping, no value function, and no need to recollect a large human demonstration dataset.

Three Exploration Enhancements

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1. Dynamic sampling

Ignore batches where all sampled trajectories either succeed or fail. This prevents zero-advantage collapse in group-relative RL.

2. Higher clipping range

Relax the PPO/GRPO upper clip boundary to allow stronger policy updates and more exploration.

3. Higher rollout temperature

Increase sampling temperature to generate more diverse trajectories during rollout.

Benchmarks and Datasets

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LIBERO

Language-guided manipulation benchmark with multiple suites:

- Spatial
- Object
- Goal
- Long
- LIBERO-90

The paper uses it to test long-horizon and generalization behavior.

RoboTwin 1.0 / 2.0

Simulation benchmark for dual-arm manipulation. RoboTwin 2.0 increases task diversity, domain randomization, and embodiment variety.

The paper describes LIBERO, RoboTwin 1.0, and RoboTwin 2.0 explicitly, and states that its RL method is applied on top of OpenVLA-OFT ?.

Why these datasets matter

- LIBERO tests language-conditioned manipulation transfer
- RoboTwin stresses longer horizons and bimanual coordination
- together they reveal whether RL improves more than one task family

Backbone used in the paper

SimpleVLA-RL is applied to **OpenVLA-OFT**, an autoregressive VLA variant based on OpenVLA with action chunking and parallel decoding.

Main Results: RL Beats SFT

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LIBERO

OpenVLA-OFT average:
91.0

After SimpleVLA-RL:
99.1

with gains across Spatial, Object, Goal, and Long suites.

RoboTwin 1.0

OpenVLA-OFT average:
39.8

After SimpleVLA-RL:
70.4

RoboTwin 2.0

OpenVLA-OFT average:
38.3

After SimpleVLA-RL:
68.8



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figures/simplevla_rl_results.png

Data Efficiency and Generalization

CV-2026

A.Kornaev,
K.Yakovlev

Modern VLA
Architecture

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Systems

Modern VLA
Post-Training
with
Reinforcement
Learning

One-trajectory setting

On LIBERO-Long, the paper reports:

17.3 → 91.7

when moving from one-trajectory SFT to RL post-training.

Generalization

The paper claims stronger:

- spatial generalization
- object generalization
- goal generalization

than SFT-only baselines.

The one-trajectory jump and the spatial/object/goal generalization claims are highlighted in the paper overview and analysis sections ?.

Interpretation

This is the central empirical message:

RL is not only optimization

RL is also capability amplification

The policy becomes more robust under sparse-data and out-of-distribution settings.

A Surprising Effect: “Pushcut”

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figures/simplevla_rl_pushcut.png

Paper Reading: SimpleVLA-RL

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SimpleVLA-RL: Scaling VLA Training via Reinforcement Learning

What to notice while reading

- Why SFT scaling is not enough
- How RL is adapted from LLM-style post-training to robot policies
- Why binary outcome rewards can be enough
- Why exploration engineering matters
- Which gains come from RL, and which come from architecture

Key message for this course

Modern VLAs are becoming **post-trained policies**. The new frontier is not only better backbones, but better **interaction-driven improvement**.

Main Takeaways of the Final Section

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Three conclusions

- 1 The 2025–2026 shift is from **imitation-only** VLA training to **SFT + RL post-training**.
- 2 Online RL can significantly improve long-horizon planning, data efficiency, and generalization.
- 3 Post-training may even induce **novel behaviors**, not only reproduce demonstrations.

Final bridge

Perception → Action → Interaction → Self-improvement

Closing remark

This is why VLA is now much closer to the broader 2026 story of foundation models: pretrain first, then improve capabilities by post-training.

Conclusions

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Modern embodied AI systems are no longer static perception models; they are adaptive decision-making systems.

Perception → Action → Interaction → Learning

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Three structural shifts

- 1 training moves from single-task data to large multimodal mixtures
- 2 policy quality improves through interaction, not only supervision
- 3 performance depends on training pipelines as much as architecture

Where the field is moving (2026)

- robot foundation models
- simulation-driven training
- RL post-training
- autonomous capability improvement

Central lesson: modern AI systems learn not only from data, but from interaction with the world.

Bibliography I

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